MSDS 535 – Week 7

Chapter 7

Overview and Objectives

- Understand feature selection and engineering.
- Compile Bayesian belief networks.
- Support vector machines.
- Select Rule-based and pattern-based classification.
- Select Classification with weak supervision.
- Select Classification with rich data type.
- Understand Potpourri: other related techniques.
- Summarize Classification advanced methods.

Filter Methods

- Filter methods select "good" features based on a goodness measure, independent of the classification model.
- Correlation between a feature and class label is a common measure, using $\chi 2$ test for categorical and Fisher score for continuous attributes.
- Fisher score considers differences in class-specific means and small class-specific variances for high scores.
- Information-theoretic measures like information gain and mutual information are also used for feature selection.
- Filter methods involve selecting k features with the highest goodness scores, building a classifier, and evaluating performance independently of the classifier's specifics, but may miss feature interactions and select redundant features.

Wrapper methods

- Wrapper methods combine feature selection and classifier training in an iterative process.
- They update the selected feature subset at each iteration based on classifier performance.
- Wrapper methods search for the best feature subset, which can be computationally expensive with exhaustive search.

Wrapper methods

- Heuristic search strategies, like stepwise forward selection and backward elimination, reduce computation.
- Some wrapper methods use advanced techniques such as simulated annealing and genetic algorithms.
- Wrapper methods tend to have better performance than filter methods but are computationally intensive.
- Embedded methods aim to combine the advantages of both filter and wrapper methods.

Embedded methods

- Embedded methods aim to combine filter and wrapper method advantages.
- They perform feature selection and classification model construction together.
- Embedded methods avoid the costly iterative search process of wrapper methods.
- Decision tree induction can be considered an embedded method as it selects features for tree nodes.
- Sparse learning techniques, like LASSO (Least Absolute Shrinkage and Selection Operator), are used for embedded methods.

Embedded methods

- LASSO minimizes a loss function penalizing the number of features used.
- Coordinate descent is an iterative algorithm for solving LASSO.
- Soft thresholding in LASSO updates feature weights to be sparse.
- Elastic net combines l1 and l2 norm regularization for feature selection in regression models.
- Logistic regression can also embed feature selection by introducing l1 norm regularization in its objective function.

Bayesian belief networks

- Bayesian belief networks allow the representation of dependencies among attributes.
- They are used for classification, and Section 7.2.1 introduces the basic concepts of these networks.
- Bayesian belief networks are also known as belief networks and probabilistic networks for brevity.
- A belief network consists of a directed acyclic graph and conditional probability tables (CPTs).
- Each node in the graph represents a random variable, and arcs represent probabilistic dependencies.

Bayesian belief networks

- A variable is conditionally independent of its nondescendants given its parents.
- CPTs specify conditional distributions for each variable based on the values of its parents.
- A belief network provides a complete representation of the joint probability distribution.
- It can have output nodes representing class labels, and various algorithms can be applied for inference and learning.
- Belief networks are used in genetic linkage analysis, computer vision, document analysis, decision support, fraud detection, and sensitivity analysis, among other applications.

Training Bayesian belief networks

- Training Bayesian belief networks involves constructing the network topology and learning conditional probability tables (CPTs).
- The network topology can be constructed by human experts or inferred from data.
- The variables in the network can be observable or hidden (missing values).
- Several algorithms exist for learning network topology from data.
- If the topology is known and variables are observable, training involves computing CPT entries.

Training Bayesian belief networks

- When some variables are hidden, gradient descent methods can be used for training.
- Gradient descent iteratively updates CPT entries to maximize an objective function.
- The algorithm computes gradients, takes steps in the gradient direction, and renormalizes weights.
- Training may benefit from prior knowledge provided by human experts.
- Belief networks offer explicit causal structure representation and can be computationally intensive.

Support vector machines

- Support Vector Machines (SVMs) are used for both linear and nonlinear data classification.
- SVMs work by transforming data into a higher-dimensional space and finding an optimal separating hyperplane.
- SVMs are known for their accuracy, ability to model complex boundaries, and resistance to overfitting.
- SVMs have been applied in various fields, including digit recognition, object recognition, emotion recognition, and more.
- Linear SVMs are explained in the context of linearly separable data.

Support vector machines

- The SVM seeks the maximum margin hyperplane (MMH) to separate classes, and support vectors are crucial for this.
- Support vectors are the closest training tuples to the MMH and define it.
- The margin represents the separation between classes, and SVM aims to minimize $\|W\|2$ to maximize the margin.
- SVMs are solved as convex quadratic programming problems, and various algorithms exist for efficient training.
- For classification, SVMs use a decision boundary equation, with αi and b determined during training.

Support vector machines

- Test tuples are classified based on the sign of the result when plugged into the decision boundary equation.
- The concept of slack variables is introduced for handling linearly inseparable data in the soft-margin linear SVM.
- Soft-margin SVM introduces a trade-off between margin size and training error, controlled by the parameter C.
- SVM complexity is determined by the number of support vectors rather than data dimensionality, making them resistant to overfitting.
- A small number of support vectors can lead to good generalization, even in high-dimensional data.

Nonlinear support vector machines

- Hard-margin and soft-margin linear SVMs are used for linearly separable and inseparable data, respectively.
- To classify linearly inseparable data, nonlinear SVMs are introduced, capable of finding nonlinear decision boundaries.
- Nonlinear SVMs involve two steps: data transformation into a higher-dimensional space and finding a linear separating hyperplane in that space.
- The choice of the nonlinear mapping and computational complexity are challenges in nonlinear SVMs.
- Kernel functions replace dot products in the training algorithm, making calculations in the original input space more efficient.

Nonlinear support vector machines

- Three admissible kernel functions are mentioned: polynomial kernel, Gaussian radial basis function kernel, and sigmoid kernel.
- SVMs can be extended for multiclass classification using strategies like training one classifier per class.
- Research focuses on improving training and testing speed for large datasets, determining the best kernel, and enhancing SVM robustness to noise.
- The kernel trick allows for nonlinear classification without explicitly constructing the nonlinear mapping.
- The kernel trick has applications beyond classification, including regression and clustering.

Rule-based and pattern-based classification

- Rule-based and pattern-based classifiers are explored in this section.
- Rule-based classifiers represent learned models as a set of IF-THEN rules.
- IF-THEN rules consist of an antecedent (IF condition) and a consequent (THEN conclusion).
- Rule-based classifiers cover tuples if their antecedents are satisfied by the tuple's attributes.
- Rules are assessed based on their coverage and accuracy.

Rule-based and pattern-based classification

- Coverage measures the percentage of tuples covered by a rule.
- Accuracy measures the percentage of covered tuples that a rule correctly classifies.
- Rule-based classification assigns class labels to tuples based on triggered rules.
- When multiple rules are triggered for a tuple, conflict resolution strategies like size ordering and rule ordering are used.
- Size ordering prioritizes rules with larger antecedents, while rule ordering organizes rules based on various criteria.

Rule-based and pattern-based classification

- A fallback or default rule can be used to assign a default class label when no rule is satisfied by a tuple.
- Default rules are evaluated if no other rule covers a tuple.
- Associative classification and discriminative frequent pattern-based classification are introduced as pattern-based classification methods.
- Associative classification uses association rules generated from frequent patterns for classification.
- Discriminative frequent pattern-based classification considers frequent patterns as combined features in addition to single features when building a classification model.

Rule extraction from a decision tree

- This subsection explores the extraction of IF-THEN rules from decision trees to create a rule-based classifier, which can be more interpretable than complex decision trees.
- Decision trees, while accurate, can become challenging to interpret as they grow in size.
- Extracting rules involves creating one rule for each path from the root to a leaf node in the decision tree, with the path's conditions becoming the rule's antecedent ("IF" part) and the leaf node holding the class prediction as the consequent ("THEN" part).

Rule extraction from a decision tree

- Pruning, a process to simplify and enhance rule interpretability, involves removing conditions from rule antecedents that do not improve rule accuracy. C4.5 uses a pessimistic approach for accuracy estimation during pruning.
- After pruning, class-based ordering is employed for conflict resolution among rules, and a final check is performed to eliminate duplicate rules. The default class selection in C4.5 prioritizes the class containing the most uncovered training tuples instead of choosing the majority class.

Rule induction using a sequential covering algorithm

- Sequential covering algorithms extract IF-THEN rules directly from training data and are widely used for mining disjunctive sets of classification rules.
- Variations of sequential covering algorithms include AQ, CN2, and RIPPER, where rules are learned one at a time, typically for one class at a time.
- Rules are created with a general-to-specific approach, starting with an empty rule and adding attribute tests to it based on heuristic criteria.
- Rule quality measures like entropy, information gain, and statistical tests are used to evaluate potential rules and their accuracy in classifying data.
- Pruning is essential to avoid overfitting, with rules being pruned based on their performance on an independent set of tuples, enhancing their generalization to unseen data.

Associative classification

- Associative classification methods, such as CBA, CMAR, and CPAR, focus on generating classification rules based on association rule mining.
- Association rule mining involves finding frequent itemsets and then generating association rules based on these itemsets, considering criteria like confidence and support.
- These classification rules are typically of the form "antecedent ⇒ class = C," where the antecedent is a conjunction of attribute–value pairs.

Associative classification

- Different associative classification methods vary in how they mine frequent itemsets, generate rules, and handle rule conflicts.
- These methods aim to create efficient classifiers that can predict class labels for new data based on the learned rules.

Discriminative frequent pattern–based classification

- Frequent patterns, derived from association rule mining, can be more discriminative than single features for classification.
- However, not all frequent patterns are equally discriminative; their power varies with their frequency and coverage.
- Frequent pattern-based classification involves generating frequent patterns as additional features alongside single features.
- Feature selection is applied to reduce the number of frequent patterns and eliminate redundant ones.

Discriminative frequent pattern–based classification

- DDPMine (Direct Discriminative Pattern Mining) directly mines discriminative patterns without generating redundant ones, using a branch-and-bound search and an FP-tree structure.
- DDPMine achieves significant speedup over the two-step approach without sacrificing classification accuracy.
- DPClass combines tree-based classifiers and feature selection to create a more compact and explainable classifier with fewer discriminative patterns, achieving similar or better performance than DDPMine.

Classification with weak supervision

- Traditional classifiers rely on a large quantity of accurately labeled training data, which can be resource-intensive.
- Real-world scenarios often present limited or entirely missing labeled data, such as in document classification, speech recognition, and computer vision.
- This chapter explores five strategies for classification in such situations:
  - Semi-supervised classification combines labeled and unlabeled data.
  - Active learning selects specific unlabeled data for human labeling.
  - Transfer learning leverages knowledge from source tasks for a new task.
  - Distant supervision automatically generates noisy labeled data from existing sources.
  - Zero-shot learning handles cases with no training data for certain class labels.
- These methods collectively fall under the category of "classification with weak supervision."

Semisupervised classification

- Semisupervised classification leverages both labeled (Xl) and unlabeled (Xu) data for building a classifier.
- Self-training involves building an initial classifier with labeled data, then iteratively labeling unlabeled data based on the classifier's predictions, potentially reinforcing errors.
- Cotraining uses multiple classifiers, each trained on different sets of features. They teach each other by labeling unlabeled data and iteratively improving their performance.
- Other approaches include modeling the joint probability distribution of features and labels and using the EM algorithm, as well as semisupervised classification with support vector machines.
- Semisupervised learning often relies on two assumptions: the clustering assumption, where data from the same cluster tend to share the same class label, and the manifold assumption, where close data points are likely to have the same class label. These assumptions guide the development of various semisupervised classification methods.

Active learning

- Active learning is an iterative form of supervised learning suitable for situations where class labels are scarce or expensive to obtain despite abundant data.
- Active learning reduces the number of labeled instances needed by purposefully querying a user, such as a human annotator, for labels.
- It starts with a small labeled subset (L) and a pool of unlabeled data (U). A querying function selects data samples from U to be labeled, and their labels are added to L.
- Active learning aims to achieve high accuracy with minimal labeled data and is evaluated using learning curves.
- Strategies for selecting which data to query include uncertainty sampling, query-by-committee, reducing the version space, and decision-theoretic approaches. Uncertainty sampling is the most common method, selecting data that the learner is least certain how to label.

Transfer learning

- Transfer learning involves applying knowledge from one or more source tasks to a target task to reduce the need for extensive labeling of data for the target task.
- It allows for adaptation to changes in data distribution and is analogous to humans applying knowledge from one task to facilitate another.
- Transfer learning is useful in applications like web-document classification and email spam filtering, where data can become outdated or distribution changes.
- The instance-based transfer learning approach, exemplified by TrAdaBoost, involves reweighting source data to learn the target task with limited labeled target data.

Transfer learning

- Negative transfer, where the transferred knowledge harms performance, is a challenge in transfer learning.
- Heterogeneous transfer learning, involving different feature spaces and multiple source domains, is an active research area.
- Transfer learning can be used with deep learning models in larger applications and is related to multitask learning, where knowledge is shared between multiple learning tasks simultaneously.

Distant supervision

- Distant supervision is a method to automatically generate labeled training data without manual annotation.
- It can be used in sentiment classification by associating positive or negative sentiment with specific symbols like ":)" or ":(" in tweets.
- Distant supervision can also leverage external knowledge sources such as web directories or YouTube video labels to automatically label data.
- It is used in tasks like social media post classification and relation extraction, but the automatically generated labels can be noisy and may not always align with the true labels.

Zero-shot learning

- Zero-shot learning aims to classify test data with class labels that were never observed during training.
- It leverages external knowledge or semantic attributes to build a classifier that can recognize novel class labels.
- The process involves training a semantic attribute classifier based on known class labels and their associated semantic attributes.
- For test data, the semantic attribute classifier predicts a semantic attribute vector, which is then compared to the attributes of novel classes to determine the class label.
- Zero-shot learning can also use class-class similarity between known and novel classes to make predictions and is applicable in various domains, including image classification and neural activity recognition.

Classification with rich data type

- Classification techniques often assume a training set with feature vectors and class labels for building classifiers.
- Different types of data beyond spatial data, such as stream data, sequence data, graph data, grid data, and spatial-temporal data, require specialized classification techniques.
- Stream data classification involves sequentially processing data as they arrive and updating the classifier with newly labeled tuples.

Classification with rich data type

- Challenges in stream data classification include high data arrival speed, the potential for an infinite stream, and concept drifting (changing class characteristics over time).
- Ensemble methods, which partition the data stream into chunks and train multiple classifiers, can address these challenges and effectively capture changing class concepts over time.
- Stream data classification has applications in finance, marketing, network monitoring, sensor networks, and other domains and can also be semi-supervised, where only a small percentage of data is labeled.

Sequence classification

- Sequence classification deals with ordered lists of values and aims to predict labels for sequences representing various data types.
- Feature engineering involves converting sequences into feature vectors, using methods like n-grams or frequent sequence patterns.
- Defining distance measures or kernel functions for sequences is an alternative approach to sequence classification.
- Some tasks involve predicting labels for each time stamp within a sequence, requiring accurate modeling of sequential dependencies.
- Traditional methods like Hidden Markov Models have limitations, while Recurrent Neural Networks offer more powerful sequence modeling capabilities.

Graph data classification

- Graph data consists of nodes connected by edges and is found in various applications like social networks, power grids, and biological networks.
- Graph data classification involves predicting labels for nodes or entire graphs, such as categorizing webpages in web graphs or identifying toxicity in molecule graphs.
- Approaches to graph classification include feature engineering, proximity measures, and deep learning techniques like Graph Neural Networks.

Graph data classification

- Random walk with restart is a useful node proximity measure for graph data, capturing relationships between nodes and paths within the graph.
- Node proximity measures can be applied to semisupervised node classification by comparing the average proximity to positively and negatively labeled nodes. Various variants and extensions of random walk with restart exist.

Potpourri: other related techniques

- Classification is a crucial data mining task with various techniques available for addressing different aspects.
- Multiclass classification extends algorithms designed for binary classification, such as support vector machines and logistic regression, to handle more than two classes.
- Approaches to multiclass classification include one-vs.-all (OVA), all-vs.-all (AVA), and error-correcting codes to improve accuracy.

Multiclass classification

- Multilabel classification involves assigning multiple labels to data tuples, which can be addressed using independent binary classifiers or label powerset transformation.
- Deep learning techniques introduced in Chapter 10 naturally handle multiclass classification by incorporating one node per class label in the output layer.

Distance metric learning

- Distance metric learning is used to automatically find the best distance metrics for a classification task.
- Mahalanobis distance, a flexible distance metric, assigns different weights to dimensions and considers their interaction effects.
- Distance metric learning aims to learn an optimal Mahalanobis distance matrix for a classification task.
- The objective is to make distances between similar data tuples small and distances between dissimilar tuples large.
- Linear and nonlinear distance metric learning methods exist, with applications in various data mining tasks beyond standard classification.

Interpretability of classification

- Interpretability is crucial in classification to help users understand the results.
- Various classification models offer different levels of interpretability (e.g., decision trees, linear classifiers).
- Model-agnostic interpretation methods like LIME use surrogate models to approximate black-box classification models locally.
- The trade-off between local fidelity and model complexity is important in interpretation.

Interpretability of classification

- Alternative approaches include counterfactual explanations and influence function analysis.
- Interpretability is essential in various data mining tasks beyond classification, such as clustering, outlier detection, ranking, and recommendation.
- Interpretability is linked to other important aspects of data mining, such as fairness, robustness, causality, and privacy.

Genetic algorithms

- Genetic algorithms emulate natural evolution in machine learning.
- Initial population consists of randomly generated rules encoded as bit strings.
- Fitness of rules is evaluated based on classification accuracy on training data.
- New populations are created by selecting fittest rules and generating offspring through crossover and mutation.
- The process continues until a population is achieved where all rules meet a specified fitness threshold.

Reinforcement learning

- Reinforcement learning helps make decisions in scenarios with uncertain rewards.
- It involves choosing actions (e.g., advertising a product) to maximize expected rewards.
- Actions have unknown values (expected rewards), and the goal is to learn the best actions.
- Strategies like ε-greedy balance exploration (trying different actions) and exploitation (choosing the best-known action).
- Reinforcement learning is used in various domains, including online advertising and robotics.

Summary

- Feature selection and engineering enhance feature relevance and construct new features.
- Bayesian belief networks capture class conditional dependencies and causal relationships.
- Support vector machines transform data into higher dimensions to find classification hyperplanes.
- Rule-based classifiers use IF-THEN rules derived from decision trees or generated directly.

Summary

- Associative classification and discriminant frequent pattern methods use frequent patterns for classification.
- Semisupervised learning combines labeled and unlabeled data with methods like self-training.
- Active learning queries for labels when data has limited or costly labels.
- Transfer learning extracts knowledge from source tasks and applies it to new tasks.
- Distant supervision generates labeled data from external information.

Summary

- Zero-shot learning predicts unseen class labels, often using semantic attributes.
- Stream data classification handles continuous data streams with scalability and concept drift challenges.
- Sequence and graph data classification predict labels for ordered sequences or nodes/graphs.
- Multiclass classification adapts binary classifiers and can use error-correcting codes for accuracy.